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# NATIONAL DEVELOPMENT COMPLEX

## Deputy Directorate General of Quality Control

Plasma Smelting Laboratory-3-H

### TEST REPORT

Test Report No: TR (027)/PS/22

Job Card No: JC (044)/PS/22

Date: 27-04-22

| INDENTER'S DATA      |                      |
|----------------------|----------------------|
| Name: Fraz-ur-Rehman | Designation: Manager |
| Group & Site: MP/3C  | Directorate: ADP     |
| Organization: NDC    | Contact No: 3627     |

| GENERAL DATA                      |                                  |                     |
|-----------------------------------|----------------------------------|---------------------|
| Sample Name: Silica Phenolic      | ID No: PP-10A-AbI-05 (Sample 05) |                     |
| Lot No: SI (820)-10A/1120-221-333 | Lot Size: 8000 RM                |                     |
| Quantity: 01                      | Chemical Composition: Composite  |                     |
| Test Type: General                | Sample Receiving Date: 26-04-22  | Test Date: 26-04-22 |

| TEST DATA                                                                            |                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Name: Ablation Test                                                             |                                                                                                                                                                                                                                                                                                                                               |
| Qualification Standard: NDC-AbI/QC/SI-C Ablatives/02 (Rev#1)                         |                                                                                                                                                                                                                                                                                                                                               |
| Standard Test Method: 1-E 458-72(Re approved 2002)                                   |                                                                                                                                                                                                                                                                                                                                               |
| Test Environmental Conditions: Ambient                                               |                                                                                                                                                                                                                                                                                                                                               |
| Test/Calorimetry Equipment & Instruments Used: Huels Arc Heater, Digital Micrometer. |                                                                                                                                                                                                                                                                                                                                               |
| Radiation & Optical Equipment used: Nil                                              | Enthalpy Measurement Technique: By Empirical Formula                                                                                                                                                                                                                                                                                          |
| Flow Temperature:                                                                    | 2952 K                                                                                                                                                                                                                                                                                                                                        |
| Flow Enthalpy:                                                                       | 3.468 MJ/Kg                                                                                                                                                                                                                                                                                                                                   |
| Duration of Exposure:                                                                | 20 Sec                                                                                                                                                                                                                                                                                                                                        |
| Results:                                                                             | ▪ Test results attached (at page 02 of 02)                                                                                                                                                                                                                                                                                                    |
| Remarks:                                                                             | ▪ These results relates only to the sample mentioned and tested above.<br>▪ This test report shall not be reproduced except in full, without written approval of the laboratory.<br>▪ The working medium for generation of Plasma Arc is air.<br>▪ Uncertainty in ablation rate calculation is $\pm 0.00774$ mm/Sec at 95 % confidence level. |

|                                                  |         |
|--------------------------------------------------|---------|
| Test Performed By: Wasim Ahmed Zahid, GM (Elect) | 27.4.22 |
| Reviewed By: Muhammad Zahid Irfan, PD (PS)       | 27.4.22 |
| Contact No: 3205/3287                            |         |

|                                     |         |
|-------------------------------------|---------|
| Approved By: Khalid Atta, Dir (STE) | 27.4.22 |
|-------------------------------------|---------|

|                    |                                   |                         |
|--------------------|-----------------------------------|-------------------------|
| Prepared By: FM/GM | Reviewed & Recommended By: GM/HOD | Approved By: GMR/HOD/PD |
|--------------------|-----------------------------------|-------------------------|

PS/FM/02, Issue 20(09-07-21)



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## Test Results

| Sr # | Description                            | Measured Value | Qualification Criteria Standard Value | Status    |
|------|----------------------------------------|----------------|---------------------------------------|-----------|
| 1    | Average initial thickness before test  | 10.935 mm      | NA                                    | Qualified |
| 2    | Final residual thickness after test    | 10.895 mm      | NA                                    |           |
| 3    | Maximum ablation in terms of thickness | 0.040 mm       | NA                                    |           |
| 4    | Ablation Rate in terms of thickness    | 0.002 mm/Sec   | $\leq 0.15$ mm / Sec                  |           |
| 5    | Max Back Surface Temperature           | NA             | NA                                    |           |

Additional Remarks: Nil

A.V. 0.002

Test Performed By:

Wasim Ahmed Zahid, GM (Elect)

End of Report

|                    |                                   |                         |
|--------------------|-----------------------------------|-------------------------|
| Prepared By: FM/GM | Reviewed & Recommended By: GM/HOD | Approved By: GMR/HOD/PD |
|--------------------|-----------------------------------|-------------------------|